

Result:
81%
26 / 32


121910 000031

Neptun ID:
Name:

Draft papers: ☐

The test consists of **12 questions**. The available time to solve the quiz is **100 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- 1.) (essay)

2.) (essay)

3.) (essay)

4.)

	a	b	c	d	e
	☐	☐	☐	☐	☐

5.)

	☐	☐	☐	☐	☐
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6.)

	☐	☐	☐	☐	☐
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7.)

	☐	☐	☐	☐	☐
--	--------------------------	--------------------------	--------------------------	--------------------------	--------------------------

8.)

	☐	☐	☐	☐	☐
	a	b	c	d	e
	☐	☐	☐	☐	☐

9.)

	☐	☐	☐	☐	☐
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10.) (essay)

11.) (essay)

12.) (essay)
- Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.
- 1.) Define the semantic value of $(A \supset B)$!

Score: 2 / 2.
- 2.) Define the atomic formula!

Score: 2 / 2.
- 3.) When can we say that a formula is satisfiable?

Score: 2 / 2.
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page 1
- ① When can we say that a formula is valid? $0 \models A$. Where A is a formula of zero-order language.

② " a formula is SAT? When its single set has a model.

③ " a set of formula is SAT? When its SAT.

④ " unSAT? When its not SAT.

⑤ Which symbols are the logical constants of zero-order language? $\wedge, \vee, \equiv, \supset, \neg$

⑥ Define the semantic equivalence of two formulas. $A \models B$ and $B \models A$.

⑦ " semantic value of $(A \supset B)$, $(A \wedge B)$, $(A \vee B)$, $(A \equiv B)$.

A	B	$A \supset B$	$A \wedge B$	$A \vee B$	$A \equiv B$
0	0	1	0	0	1
0	1	1	0	1	0
1	0	0	0	1	0
1	1	1	1	1	1

⑧ " atomic formula. Non logical constants are atomic formula.

⑨ " interpretation of zero-order language. $\mathcal{S} \in \{0,1\}^{Con}$ is the interpretation of zero-....

⑩ " CNF. elementary disjunction.

⑪ " elementary conjunction. a literal

⑫ " literal. if $P \in Con$, then P and $\neg P$ formulas are literal.

⑬ Formula labeled the root of construction tree. atomic formula.
- 4.) Which interpretations are **models** of the following set of formulae?
- $$\{ \neg(p \equiv q), ((p \supset (q \equiv q)) \supset q) \}$$
- (a) $\varrho(p) = 0, \varrho(q) = 0$

2 $\varrho(p) = 0, \varrho(q) = 1$

(c) $\varrho(p) = 1, \varrho(q) = 0$

(d) $\varrho(p) = 1, \varrho(q) = 1$
- 5.) Which interpretations are models of the following set of formulae?
- $$\{ ((q \supset q) \vee (q \vee p)), (p \supset \neg q) \}$$
- 2** $\varrho(p) = 0, \varrho(q) = 0$

2 $\varrho(p) = 0, \varrho(q) = 1$

2 $\varrho(p) = 1, \varrho(q) = 0$

(d) $\varrho(p) = 1, \varrho(q) = 1$
- 6.) Which interpretations are models of the following set of formulae?
- $$\{ ((q \equiv q) \supset p), ((q \supset p) \equiv \neg q) \}$$
- (a) $\varrho(p) = 0, \varrho(q) = 0$

(b) $\varrho(p) = 0, \varrho(q) = 1$

2 $\varrho(p) = 1, \varrho(q) = 0$

(d) $\varrho(p) = 1, \varrho(q) = 1$
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- | P | q | $\neg(p \equiv q)$ | | $((p \supset (q \equiv q)) \supset q)$ | | |
|---|---|--------------------|---|--|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 1 | 1 | 1 | 1 |
- | P | q | $((q \supset q) \vee (q \vee p))$ | | | $(p \supset \neg q)$ | |
|---|---|-----------------------------------|---|---|----------------------|---|
| 0 | 0 | 1 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 1 | 0 |
| 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 |
- | P | q | $((q \equiv q) \supset p)$ | | $((q \supset p) \equiv \neg q)$ | | |
|---|---|----------------------------|---|---------------------------------|---|---|
| 0 | 0 | 1 | 0 | | | |
| 0 | 1 | 1 | 0 | | | |
| 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 |

7.) Which statements are **true** for the following three formula?

$$A \models q \wedge q, \quad B \models \neg p, \quad C \models \neg(p \equiv q) \equiv q \supset q$$

- (a) $A \Leftrightarrow B$,
2 $\{A, B, C\}$ is satisfiable,
2 $A, B \models C$,
 (d) $(A \vee B)$ is valid.

a) $A \Leftrightarrow B : A \models B, B \models A : \{A, \neg B\}, \{B, \neg A\}$
 ① $\{(q \wedge q), \neg(\neg p)\}$ it has model $\rightarrow$ SAT

p	q	$q \wedge q$	$\neg(\neg p)$
0	0	0	0
0	1	1	0
1	0	0	1
1	1	1	1

b)

A	B	$(\neg(p \equiv q)) \equiv (q \supset q)$
0	1	0
1	1	1
0	0	1
1	0	0

c)

A	B	$\neg C$
0	1	1
1	1	0
0	0	0
1	0	1

d)

A	B
0	1
1	1
0	0
1	0

8.) Let $L^{(1)}$ be a first-order language defined as follows:

$$L^{(1)} = \langle LC, \{x_1, x_2, \dots\}, \{P, Q, c\}, Term, Form \rangle$$

$$P \in \mathcal{P}(1), \quad Q \in \mathcal{P}(2), \quad c \in \mathcal{F}(0).$$

Let $\langle U, \varrho \rangle$ be an interpretation for $L^{(1)}$ such that:

- $U = \{u_1, u_2, u_3\}$
- $\varrho(P) = P' \quad P'(x) = \begin{cases} 1 & \text{if } x = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(Q) = Q' \quad Q'(x, y) = \begin{cases} 1 & \text{if } x = u_1 \text{ and } y = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(c) = u_2$

Let v be an assignment relying on the interpretation such that $v : Var \rightarrow U$

$$v(x_1) = u_3, \quad v(x_2) = u_1$$

Which formulas are **true** (1) from the ones below?

- 1** $\neg \exists x_1 Q(x_1, x_1)$
0 $\neg \forall x_2 Q(x_2, x_2)$
-1 $\forall x_1 \neg P(x_1)$
0 $\exists x_2 \neg P(x_2)$
 (e) $\neg \forall x_2 \neg Q(x_2, x_1)$

$\exists$ all 1 are true

$\forall$ all 1 are true

a) $\neg \exists x_1 Q(x_1, x_1)$
 0 1 $Q(u_1, u_1) = 1$ False
 $Q(u_2, u_2) = 0$
 $Q(u_3, u_3) = 0$

b) $\neg \forall x_2 Q(x_2, x_2)$
 1 0 $Q(u_1, u_1) = 1$ True
 $Q(u_2, u_2) = 0$
 $Q(u_3, u_3) = 0$

c) $\forall x_1 \neg P(x_1)$
 0 0 $P(u_1) = 1$ False
 1 $P(u_2) = 0$
 1 $P(u_3) = 0$

d) $\exists x_2 \neg P(x_2)$ True
 0 $P(u_1) = 1$
 1 $P(u_2) = 0$
 1 $P(u_3) = 0$

e) $\neg \forall x_2 \neg Q(x_2, x_1)$
 0 1 1 $Q(u_1, u_2) = 0$ False
 1 $Q(u_1, u_3) = 0$
 1 $Q(u_2, u_3) = 0$

9.) How can we formalize the following statement by using a zero-order language with $Con = \{p, q, r\}$?

$(r \wedge p) \supset \neg q$
If it is daytime and the sky is blue then there are no clouds.

$p \approx$ The sky is blue. $q \approx$ There are clouds on the sky. $r \approx$ It is daytime.

(a) $\neg q \supset (r \wedge p)$

2 (b) $(r \wedge p) \supset \neg q$

(c) $\neg q \equiv \neg r \wedge p$

(d) $(r \wedge p) \equiv \neg q$

10.) What is the conjunctive normal form of $((p \supset q) \equiv r)$?

Score: 2 / 2.

$(\neg q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee r)$

11.) Construct the prenex form of

$(\exists x (Q(x, y) \supset \neg \forall x (Q(x, z) \vee P(z))) \supset \forall x P(x))$.

Score: 2 / 2.

12.) Prove the following with sequent calculus.

$(p \wedge q) \vee \neg r \models (r \supset p) \wedge (r \supset q)$

Score: 2 / 2.

Deduction rules of Gentzen calculus

$(\Rightarrow)$ $\frac{A, \Gamma \rightarrow \Delta, B; B, \Gamma \rightarrow \Delta, A}{\Gamma \rightarrow \Delta, (A \equiv B)}$ $(\Rightarrow)$ $\frac{A, B, \Gamma \rightarrow \Delta; \Gamma \rightarrow \Delta, A, B}{(A \equiv B), \Gamma \rightarrow \Delta}$ $(A \equiv B) \text{ is } \text{iff } A \text{ and } B \text{ are equivalent}$

$(\supset)$ $\frac{A, \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \supset B)}$ $(\supset)$ $\frac{\Gamma \rightarrow \Delta, A; B, \Gamma \rightarrow \Delta}{(A \supset B), \Gamma \rightarrow \Delta}$

$(\wedge)$ $\frac{\Gamma \rightarrow \Delta, A; \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \wedge B)}$ $(\wedge)$ $\frac{A, B, \Gamma \rightarrow \Delta}{(A \wedge B), \Gamma \rightarrow \Delta}$

$(\vee)$ $\frac{\Gamma \rightarrow \Delta, A; \Gamma \rightarrow \Delta, B}{\Gamma \rightarrow \Delta, (A \vee B)}$ $(\vee)$ $\frac{A, \Gamma \rightarrow \Delta; B, \Gamma \rightarrow \Delta}{(A \vee B), \Gamma \rightarrow \Delta}$ $(A \vee B) \text{ is } \text{iff } A \text{ or } B \text{ is true}$

$(\neg)$ $\frac{A, \Gamma \rightarrow \Delta}{\Gamma \rightarrow \Delta, \neg A}$ $(\neg)$ $\frac{\Gamma \rightarrow \Delta, A}{\neg A, \Gamma \rightarrow \Delta}$

⑩ $p \quad q \quad r \quad ((p \supset q) \equiv r)$

0	0	0	1	0
0	0	1	1	1
0	1	0	1	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	0
1	1	0	1	0
1	1	1	1	1

$(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$

	q	$\neg q$
p	$\boxed{0} \quad \boxed{1}$	$\boxed{0} \quad \boxed{1}$
$\neg p$	$\boxed{0} \quad \boxed{1}$	$\boxed{1} \quad \boxed{0}$

$\neg r$ r $\neg r$

$\neg ((q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg r)) = (\neg q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee r)$

⑪

$(\exists x_1 (Q(x_1, y) \supset \neg \forall x_2 (Q(x_2, z) \vee P(z))) \supset \forall x_3 P(x_3))$

$(\exists x_1 (Q(x_1, y) \supset \neg \forall x_2 (Q(x_2, z) \vee P(z))) \supset \forall x_3 P(x_3))$

$\forall x_1 \forall x_2 \forall x_3 (Q(x_1, y) \supset \neg (Q(x_2, z) \vee P(z))) \supset P(x_3)$

⑫ $(p \wedge q) \vee \neg r \models (r \supset p) \wedge (r \supset q)$

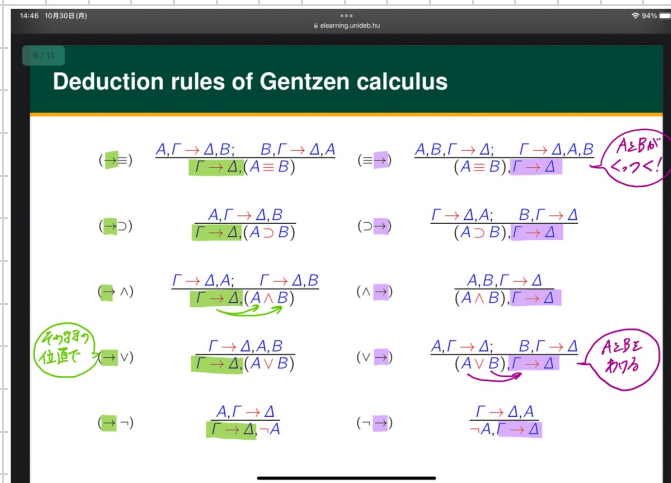
$\frac{r, p \rightarrow p}{p, q \rightarrow (r \supset p)} ; \frac{r, p \rightarrow q}{p, q \rightarrow (r \supset q)} \quad \frac{r \rightarrow r \supset p}{\neg r \rightarrow (r \supset p)} ; \frac{r \rightarrow r \supset q}{\neg r \rightarrow (r \supset q)} \quad (\rightarrow \supset)$

$\frac{(p \wedge q) \rightarrow (r \supset p); (p \wedge q) \rightarrow (r \supset q)}{(\rightarrow \wedge) (p \wedge q) \rightarrow (r \supset p) \wedge (r \supset q)} ; \frac{\neg r \rightarrow (r \supset p) \wedge (r \supset q)}{(\rightarrow \wedge) (p \wedge q) \rightarrow (r \supset p) \wedge (r \supset q)}$

$(\vee \rightarrow) (p \wedge q) \vee \neg r \rightarrow (r \supset p) \wedge (r \supset q)$

Practice - sequent calculus

$$\begin{array}{lcl}
 \text{ax.} & & \text{ax.} \\
 \textcircled{P} \rightarrow \textcircled{P} \neg q, \neg q \ ; \ \textcircled{\neg q} \rightarrow \textcircled{P}, \textcircled{\neg q}, P & & \\
 \hline
 \rightarrow P, (P \equiv \neg q), \neg q & & (\rightarrow \equiv) \\
 \hline
 \neg P \rightarrow (P \equiv \neg q), \neg q & & \\
 \hline
 \rightarrow (P \equiv \neg q), (\neg P \supset \neg q) & & (\rightarrow \supset) \\
 \hline
 \neg (P \equiv \neg q) \rightarrow (\neg P \supset \neg q) & & \\
 \hline
 \rightarrow \neg (P \equiv \neg q) \supset (\neg P \supset \neg q) & & (\rightarrow \supset)
 \end{array}$$



$$\begin{array}{l}
 \text{ax.} \\
 \frac{q, p, r, p \rightarrow q}{p, r, p \rightarrow q, \neg q; \neg q, r, p \rightarrow q, p} \quad \text{ax.} \quad \frac{r, p \rightarrow r (p \equiv \neg q)}{\neg r, r, p \rightarrow r (p \equiv \neg q)} \quad \text{ax.} \\
 \frac{(\rightarrow \Rightarrow) \quad r, p \rightarrow (p \equiv \neg q), q; \neg r, r, p \rightarrow (p \equiv \neg q)}{p, q \supset \neg r \rightarrow (p \equiv \neg q), p; r, p, q \supset \neg r \rightarrow (p \equiv \neg q)} \quad (\supset \rightarrow) \\
 \frac{p, p \supset r, q \supset \neg r \rightarrow (p \equiv \neg q)}{\neg (p \equiv \neg q), p \supset r, q \supset \neg r \rightarrow \neg p} \quad (\supset \rightarrow)
 \end{array}$$

ax.	$\underline{P, Q \rightarrow r, Q}; (R, P, Q \rightarrow R)$	ax.	$\underline{Q, P \rightarrow r, P}; (R, Q, P \rightarrow R)$
$(\supset \rightarrow)$	$\underline{P, Q, P \supset (Q \rightarrow r)} \rightarrow r$	$(\supset \rightarrow)$	$\underline{Q, P, Q \supset (P \rightarrow r)} \rightarrow r$
$(\supset \supset)$	$\underline{Q, P \supset (Q \rightarrow r)} \rightarrow (P \supset r)$	$(\supset \supset)$	$\underline{P, Q \supset (P \rightarrow r)} \rightarrow (Q \supset r)$
$(\supset \supset)$	$\underline{P \supset (Q \rightarrow r)} \rightarrow Q \supset (P \rightarrow r);$ $\rightarrow P \supset Q \supset r \equiv Q \supset P \supset r$	$(\supset \supset)$	

$$\begin{array}{c} \text{ax.} \\ \frac{P, Q \rightarrow Q, r}{P, Q \rightarrow (Q \vee r)} \\ \frac{(P \wedge Q) \rightarrow P ; (P \wedge Q) \rightarrow (Q \vee r)}{(\rightarrow \wedge) \frac{(P \wedge Q) \rightarrow P \wedge (Q \vee r)}{(P \wedge Q) \vee (P \wedge r)} ; (P \wedge r) \rightarrow P ; (P \wedge r) \rightarrow (Q \vee r)} \\ \frac{(\rightarrow \vee) \frac{(P \wedge Q) \vee (P \wedge r) \rightarrow P \wedge (Q \vee r)}{\neg(P \wedge (Q \vee r)), (P \wedge Q) \vee (P \wedge r) \rightarrow} \end{array}$$

$$\textcircled{1} \quad \left(\exists x_1 P(x_1) \supset \neg \left(\left(\forall x_2 Q(x_2, y) \supset \neg \exists x_3 P(x_3) \right) \wedge \left(\neg \exists x_4 P(x_4) \supset \forall x_5 Q(x_5, y) \right) \right) \right)$$

$$\forall x_1 \forall x_2 \exists x_3 \forall x_4 \exists x_5 (\text{---})$$

$$\textcircled{2} \quad \left(\exists x_1 (Q(x_1, y) \supset \neg \forall x_2 (Q(x_2, z) \vee P(z))) \supset \forall x_3 P(x_3) \right)$$

$$\forall x_1 \forall x_2 \forall x_3 (\text{---})$$

$$\textcircled{3} \quad \left(\forall x_1 P(x_1) \supset \exists x_2 (Q(x_2, y) \supset \neg \forall x_3 (Q(x_3, z) \vee P(z))) \right)$$

$$\exists x_1 \exists x_2 \exists x_3 (\text{---})$$

$$\textcircled{4} \quad \left(\exists x_1 (Q(x_1, y) \supset \neg (\neg \forall x_2 Q(x_2, z) \supset P(z))) \supset \forall x_3 P(x_3) \right)$$

$$\forall x_1 \forall x_2 \forall x_3 (\text{---})$$

$$\textcircled{5} \quad \left(\forall x_1 (Q(x_1, y) \supset \neg (\neg \exists x_2 Q(x_2, z) \supset P(z))) \supset \exists x_3 P(x_3) \right)$$

$$\exists x_1 \exists x_2 \exists x_3 (\text{---})$$

$$\exists x A(x) = \neg \forall x \neg A(x)$$

$$(A = B) \Leftrightarrow (A \supset B) \wedge (B \supset A)$$

$$\left(\underbrace{\exists x_1 A(x_1)}_{x_1} \supset \underbrace{\forall x_2 A(x_2)}_{x_2} \right) \wedge \left(\underbrace{\forall x_3 A(x_3)}_{x_3} \supset \underbrace{\exists x_4 A(x_4)}_{x_4} \right)$$

$$\left(\exists x_1 A(x_1) \supset \forall x_2 A(x_2) \right) \wedge \left(\forall x_3 A(x_3) \supset \exists x_4 A(x_4) \right)$$

$$\forall x_1 \forall x_2 \exists x_3 \exists x_4 \left((A(x_1) \supset A(x_2)) \wedge (A(x_3) \supset A(x_4)) \right)$$

Result:
91%
29 / 32Neptun ID: Draft papers: ☐Name:

The test consists of **12 questions**. The available time to solve the quiz is **100 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- 1.) (essay)
2.) (essay)
3.) (essay)

	a	b	c	d	e
4.)	☐	☐	☐	☐	☐
5.)	☐	☐	☐	☐	☐
6.)	☐	☐	☐	☐	☐
7.)	☐	☐	☐	☐	☐
8.)	☐	☐	☐	☐	☐

	a	b	c	d	e
9.)	☐	☐	☐	☐	☐
10.)	☐	☐	☐	☐	☐
11.)	☐	☐	☐	☐	☐
12.)	☐	☐	☐	☐	☐

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) When can we say that a set of formulae is unsatisfiable?

Score: 2 / 2.

2.) Define the semantic value of $(A \wedge B)$!

Score: 2 / 2.

3.) When can we say that a formula is valid?

Score: 2 / 2.

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4.) Which interpretations are **models** of the following set of formulae?

$$\{ (q \supset (q \vee q)), \neg((q \equiv p) \supset p) \}$$

☒ $\varphi(p) = 0, \varphi(q) = 0$

(b) $\varphi(p) = 0, \varphi(q) = 1$

(c) $\varphi(p) = 1, \varphi(q) = 0$

(d) $\varphi(p) = 1, \varphi(q) = 1$

5.) Which interpretations are **models** of the following set of formulae?

$$\{ (q \supset ((q \vee p) \equiv q)), \neg(q \supset p) \}$$

(a) $\varphi(p) = 0, \varphi(q) = 0$

☒ $\varphi(p) = 0, \varphi(q) = 1$

(c) $\varphi(p) = 1, \varphi(q) = 0$

(d) $\varphi(p) = 1, \varphi(q) = 1$

6.) Which interpretations are **models** of the following set of formulae?

$$\{ ((q \supset q) \vee (q \vee p)), (p \supset \neg q) \}$$

☒ $\varphi(p) = 0, \varphi(q) = 0$

☒ $\varphi(p) = 0, \varphi(q) = 1$

☒ $\varphi(p) = 1, \varphi(q) = 0$

(d) $\varphi(p) = 1, \varphi(q) = 1$

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but
A

P	q	$(q \supset (q \vee q))$	$\neg((q \equiv p) \supset p)$
0	0	1	0
0	1	1	0
1	0	1	0
1	1	1	1

P	q	$(q \supset ((q \vee p) \equiv q))$	$\neg(q \supset p)$
0	0	1	1
0	1	1	0
1	0	1	1
1	1	1	0

P	q	$((q \supset q) \vee (q \vee p))$	$(p \supset \neg q)$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	1	0

7.) Which statements are **true** for the following three formula?

$$A \equiv p \supset p, \quad B \equiv \neg(q \supset (q \equiv q)), \quad C \equiv \neg q \equiv q$$

- (a) $A \Leftrightarrow B$,
 (b) $\{A, B, C\}$ is satisfiable,

☒ $A, B \equiv C$,

☒ $(A \vee B)$ is valid.
8.) Let $L^{(1)}$ be a first-order language defined as follows:

$$L^{(1)} = \{LC, \{x_1, x_2, \dots\}, \{P, Q, c\}, Term, Form\}$$

$$P \in \mathcal{P}(1), \quad Q \in \mathcal{P}(2), \quad c \in \mathcal{F}(0).$$

Let $\langle U, \varrho \rangle$ be an interpretation for $L^{(1)}$ such that:

- $U = \{u_1, u_2, u_3\}$
- $\varrho(P) = P'$ $P'(x) = \begin{cases} 1 & \text{if } x = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(Q) = Q'$ $Q'(x, y) = \begin{cases} 1 & \text{if } x = u_1 \text{ and } y = u_1 \\ 0 & \text{otherwise} \end{cases}$
- $\varrho(c) = u_2$

Let v be an assignment relying on the interpretation such that $v: Var \rightarrow U$

$$v(x_1) = u_3, \quad v(x_2) = u_1$$

Which formulas are **true** (1) from the ones below?

- (a) $\neg \exists x_1 P(x_1)$
 (b) $\neg \forall x_2 \neg Q(x_2, x_1)$
 (c) $\forall x_1 \neg Q(x_1, x_1)$
- ☒ $\exists x_2 \neg Q(x_2, x_2)$
☒ $\neg \forall x_2 P(x_2)$

a) $A \Leftrightarrow B : A \models B, B \models A$ ① $A \models B$ it has model $\rightarrow$ SAT

P	q	$p \supset p$	$\neg(\neg(q \supset (q \equiv q)))$
0	0	1	1 0
0	1	1	1 0
1	0	1	1 0
1	1	1	1 0

b) $A \models B, \neg q \equiv q$

A	B	$\neg q \equiv q$
1	0	
1	0	
1	0	
1	0	

it has no model $\rightarrow$ unSAT

c) $A \models B, \neg(\neg q \equiv q)$

A	B	$\neg(\neg q \equiv q)$
1	0	1 0
1	0	1 0 0
1	0	1 1 0
1	0	1 0 0

it has no model $\rightarrow$ unSAT $\rightarrow A, B \models C$

d) $A \models B$

A	B
1	0 = 1
1	0 = 1
1	0 = 1
1	0 = 1

Valid

a) $\neg \exists x_1 P(x_1)$

$0 \leftarrow 1$ $P(u_1) = 1$ False
 $P(u_2) = 0$
 $P(u_3) = 0$

b) $\neg \forall x_2 \neg Q(x_2, x_1)$

$0 \leftarrow 1$ $1 Q(u_1, u_2) = 0$ False
 $1 Q(u_1, u_3) = 0$
 $1 Q(u_2, u_3) = 0$

c) $\forall x_1 \neg Q(x_1, x_1)$

0 $0 Q(u_1, u_1) = 1$ False
 $1 Q(u_2, u_2) = 0$
 $1 Q(u_3, u_3) = 0$

d) $\exists x_2 \neg Q(x_2, x_2)$

1 $0 Q(u_1, u_1) = 1$ True
 $1 Q(u_2, u_2) = 0$
 $1 Q(u_3, u_3) = 0$

e) $\neg \forall x_2 P(x_2)$

$1 \leftarrow 0$ $P(u_1) = 1$ True
 $P(u_2) = 0$
 $P(u_3) = 0$

9.) How can we formalize the following statement by using a zero-order language with $Con = \{p, q, r\}$?

$(r \wedge p) \supset \neg q$
if it is daytime and the sky is blue then there are no clouds.

$p \approx$ The sky is blue. $q \approx$ There are clouds on the sky. $r \approx$ It is daytime.

(a) $(r \wedge p) \equiv \neg q$

☒ -1 $\neg q \supset (r \wedge p)$

☐ 0 $(r \wedge p) \supset \neg q$

(d) $\neg q \equiv \neg r \wedge p$

10.) What is the conjunctive normal form of $((p \supset q) \equiv r)$?

Score: 2 / 2.

CNF: $(\neg q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee r)$

11.) Construct the prenex form of

$(\exists x (Q(x, y) \supset \neg \forall x (Q(x, z) \vee P(z))) \supset \forall x P(x))$.

Score: 2 / 2.

12.) Prove the following with sequent calculus.

$(p \wedge q) \vee \neg r \vdash (r \supset p) \wedge (r \supset q)$

Score: 2 / 2.

⑩

p	q	r	$((p \supset q) \equiv r)$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$

	q	$\neg q$
p	0	1
$\neg p$	1	0

$\neg((q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg r))$

CNF: $(\neg q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee r)$

⑪ $(\exists x_1 (Q(x_1, y) \supset \neg \forall x_2 (Q(x_2, z) \vee P(z))) \supset \forall x_3 P(x_3))$

$(\exists x_1 (Q(x_1, y) \supset \neg \forall x_2 (Q(x_2, z) \vee P(z))) \supset \forall x_3 P(x_3))$

$\forall x_1 \forall x_2 \forall x_3 ((Q(x_1, y) \supset \neg (Q(x_2, z) \vee P(z))) \supset P(x_3))$

⑫

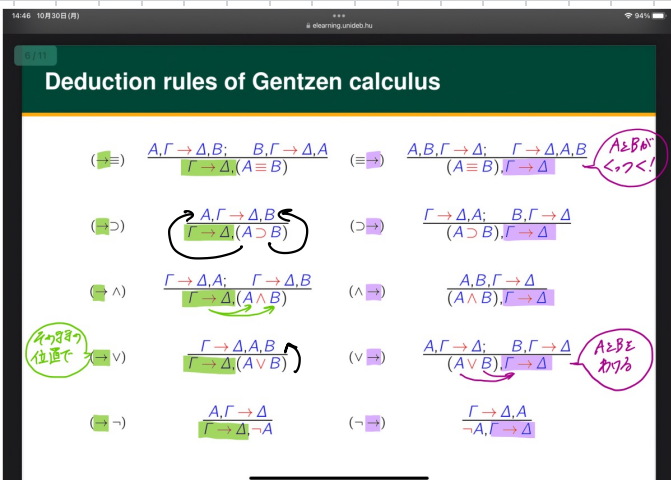
ax. $r, p, q \rightarrow p$ ax. $r, p, q \rightarrow q$ ax. $r \rightarrow r \vee p$ ax. $p \rightarrow p \vee q$

$(\rightarrow \supset)$ $p, q \rightarrow (r \vee p); p, q \rightarrow (r \vee q) \rightarrow r, (r \vee p); \rightarrow r, (r \vee q)$ $(\rightarrow \supset)$

$(\wedge \rightarrow)$ $(p \wedge q) \rightarrow (r \vee p); (p \wedge q) \rightarrow (r \vee q) \rightarrow r, (r \vee p) \wedge (r \vee q)$ $(\rightarrow \wedge)$

$(\rightarrow \wedge)$ $(p \wedge q) \rightarrow (r \vee p) \wedge (r \vee q); \neg r \rightarrow (r \vee p) \wedge (r \vee q)$

$(p \wedge q) \vee \neg r \rightarrow (r \vee p) \wedge (r \vee q)$ $(\vee \rightarrow)$



$(\rightarrow \equiv)$ $\frac{A \vdash \Delta, B; B \vdash \Delta, A}{\vdash \Delta, (A \equiv B)}$	$(\equiv \rightarrow)$ $\frac{A, B \vdash \Delta; \vdash \Delta, A, B}{(A \equiv B), \vdash \Delta}$
$(\rightarrow \supset)$ $\frac{A \vdash \Delta, B}{\vdash \Delta, (A \supset B)}$	$(\supset \rightarrow)$ $\frac{\vdash \Delta, A; B \vdash \Delta}{(A \supset B), \vdash \Delta}$
$(\rightarrow \wedge)$ $\frac{\vdash \Delta, A; \vdash \Delta, B}{\vdash \Delta, (A \wedge B)}$	$(\wedge \rightarrow)$ $\frac{A, B \vdash \Delta}{(A \wedge B), \vdash \Delta}$
$(\rightarrow \vee)$ $\frac{\vdash \Delta, A; \vdash \Delta, B}{\vdash \Delta, (A \vee B)}$	$(\vee \rightarrow)$ $\frac{A \vdash \Delta; B \vdash \Delta}{(A \vee B), \vdash \Delta}$
$(\rightarrow \neg)$ $\frac{A \vdash \Delta}{\vdash \Delta, \neg A}$	$(\neg \rightarrow)$ $\frac{\vdash \Delta, A}{\neg A, \vdash \Delta}$